

Matrix multiplication of columns times rows instead of rows times columns

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In ordinary matrix multiplication AB where we multiply each column b_i by A , each resulting column of AB can be viewed as a linear combination of A .

If however if we decided to multiply each column of A by each row of B , we get an entire matrix for each column-row multiply. My question is: Does each matrix resulting from an outer product have any known meaning aside from being a part of the sum(a summand?) of the final AB ?

Edit: Say we have AB

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 15 & 16 & 17 \end{pmatrix}$$

Normally we would multiply each column of B by A and get a *linear combination of A*, e.g.

$$10 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + 13 \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 15 \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

which is one column of AB .

If however we multiply each column of A by each row of B , e.g.

$$\begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} (10 \quad 11 \quad 12)$$

we get a matrix. Each of the 3 matrices $a_i b_i^T$ summed together gives us AB . I was wondering if each individual matrix that sums to AB has any sort of special meaning. This second way of performing multiplication also seems to be called column-row expansion.

(<http://www.math.nyu.edu/~neylon/linalgfallo4/project1/dj/crexpansion.htm>). I actually read about it in I believe section 2.4 of Strang's Introduction to Linear Algebra book. He mentions that not everybody is aware that matrix multiplication can be performed in this way.

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matrices

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edited Sep 5, 2015 at 10:14

asked Sep 4, 2015 at 15:11



John

1,149

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- 1 The operation looks very similar to the tensor product, but I don't fully understand the last two sentences..
– Peter Franek Sep 4, 2015 at 19:14

1 Answer

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2



Before talking about multiplication of two matrices, let's see another way to interpret matrix A . Say we have a matrix A as below,

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

we can easily find that column $\begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$ is linear combination of first two columns.

$$1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$$

And you can say $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$ are two basis for column space of A .

Forgive the reason why you want to decompose matrix A at first place like this,

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{bmatrix}$$

but you can, and in the end, it looks reasonable.

If you view this equation column wise, each $column_j$ of A is the sum of corresponding $column_j$ of each matrix in RHS.

What's special about each matrix of RHS is that each of them is a rank 1 matrix whose column space is

the line each base of column space of A lies on. e.g. $\begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ spans only $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. And people say rank 1

matrices are the building blocks of any matrices.

If now you revisit the concept of viewing A column by column, this decomposition actually emphasizes the concept of linear combination of base vectors.

If these make sense, you could extend the RHS further,

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$$

Each term in RHS says take this base, and make it "look like" a rank 3 matrix.

And we can massage it a little bit, namely put RHS into matrix form, you get

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Now you can forget matrix A , and imagine what you have are just two matrices on RHS. When you read this text backward(I mean logically), I hope matrix multiplication in this fashion makes sense to you now. Or if you prefer, you can start with two matrices in the question.

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answered Sep 22, 2019 at 8:04



Adam

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